HDFC BANK • BOARD AND EXECUTIVE BRIEFING

Quantum-Safe Communications

PQC + QKD architecture for HDFC —
HQ Mumbai • Primary DC Mumbai • Far DR Delhi

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WHY WE ARE HERE

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Quantum-safe communications — the ask in one slide

THE THREAT

Adversaries can record encrypted traffic today and decrypt it in 2030–35 once cryptographically-relevant quantum computers (CRQC) arrive — “harvest now, decrypt later”. Our HQ–DC, DC–DR and partner-bank links all use RSA/ECDH today.

THE ARCHITECTURE

Two-layer defence — **PQC (Kyber + Dilithium)** as the workhorse on every link including branches and customer apps; **QKD as a selective overlay** on the Mumbai HQ–DC metro fibre. AES-256 inside the HSM is already largely quantum-safe.

THE ASK

Rs 50–70 Cr over FY26–FY28: HSM refresh + PQC migration + QKD pilot on the HQ–DC metro link + R&D bet on satellite-QKD with ISRO for the Delhi DR — puts HDFC ahead of RBI and on the National Quantum Mission map.

WHAT GOOD LOOKS LIKE BY FY28

100%

of internal mTLS, branch IPsec, and customer-facing TLS migrated to **hybrid PQC**

1 Gbps+

QKD-secured DWDM channel between HQ Mumbai and Primary DC Mumbai

AES-256

rotated **every ~1 sec** on the HQ–DC link via QKD-derived symmetric keys

Rs 0

regulatory penalty exposure on **HNDL** — RBI / SEBI / FIU expectations met early

WHAT WE ARE PROTECTING

HDFC topology — and the distance reality check

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FIBRE TOPOLOGY

HQ MUMBAI

Lower Parel / BKC
Treasury, FX, MIS

5–30 km
intra-Mumbai dark fibre

PRIMARY DC MUMBAI

Andheri / Navi Mumbai
Core banking, card switch,
UPI / NEFT / RTGS, fraud

~1,150 km

carrier DWDM / MPLS, multi-hop

FAR DR DELHI

Different seismic zone
Async replica
RPO few minutes

8,000+ BRANCHES • 18,000+ ATMS

MPLS / SD-WAN / 4G-5G failover — multi-hop carrier IP, no dark fibre

DISTANCE REALITY CHECK

HQ Mumbai ↔ DC Mumbai

5–30 km — QKD: yes — sweet spot.

Standard Toshiba MDI-QKD or ID Quantique Clavis comfortably handle this; coexistence with classical DWDM proven up to ~100 km (JPMC 2022).

DC Mumbai ↔ DR Delhi

~1,150 km — QKD: no commercial path.

QKD is exponentially loss-limited (~0.2 dB/km). Practical reach ~120 km. Need trusted-node relays (China Beijing–Shanghai) or satellite (Micius / EAGLE-1) — not available in India yet.

Branches ↔ DC

10–2,000 km — QKD: physically infeasible.

Multi-hop carrier IP/MPLS, shared paths, no dark fibre. PQC over SD-WAN is the only viable path.

WHY THIS IS URGENT, NOT THEORETICAL

Harvest now, decrypt later — the urgent threat

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THE 10-YEAR THREAT TIMELINE

2024–26

Adversaries silently record encrypted traffic from carrier fibre, optical amplifier huts, or compromised leased lines.

2027–29

NIST PQC standards (Kyber, Dilithium) become mandatory in regulated sectors. Banks that have not migrated face audit findings.

2030–32

First **CRQCs** with 1–3 M physical qubits. Shor's algorithm runs against captured RSA-2048 / ECDH-256 handshakes.

2033–35

All **harvested 2024–32 traffic is decrypted**. KYC, loan files, treasury positions, source code, HSM key wraps — all readable.

WHAT IS ON HDFC'S HQ–DC AND DC–DR FIBRE TODAY

Core banking journal

Every CASA, FD, loan tx — PII + balances. 10s of Gbps.

Card switch

Every card auth — PAN, ARQC cryptograms. 100k+ tps peak.

UPI / NEFT / RTGS

Every retail transfer — VPA, beneficiary, amount.

HSM cluster sync

Master keys that derive every PIN, MAC, working key. Tiny but catastrophic.

Treasury / FX feeds

Live rates, derivatives positions, trade tickets.

Backups + admin sessions

Full DB images. SSH/RDP root creds.

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Two defences — and why we need both

*PQC handles **breadth** — every TLS, every IPsec, every branch. QKD handles **depth** — the 2–4 highest-value pipes inside a metro.*

POST-QUANTUM CRYPTOGRAPHY (PQC)

Math-based defence — software upgrade

- Hard-math problems believed secure against **classical + quantum**: lattice (ML-KEM Kyber), hash-based signatures (SLH-DSA SPHINCS+).
- **Drop-in replacement** for ECDH/RSA in TLS, IPsec, SSH, JWT, code signing.
- Run in **hybrid mode** — ECDH + Kyber together — so neither algorithm alone needs to be perfect.
- **NIST FIPS-203 / 204 / 205** standardised Aug 2024. Apple iOS 18, Chrome, Cloudflare ship Kyber today.
- Cost — **firmware + library upgrade**, no new hardware. Same fibre, same boxes.
- **Use everywhere:** branches, customer apps, partner APIs, internal services.

QUANTUM KEY DISTRIBUTION (QKD)

Physics-based defence — selective overlay

- Sends single photons — **measuring disturbs them** (Heisenberg uncertainty), so any eavesdropper is detected.
- Generates a stream of **fresh AES-256 keys every ~1 sec** between two endpoints over fibre.
- Feeds those keys into existing **MACsec / IPsec / OTN encryptors** for AES-256 bulk encryption of 10–400 Gbps.
- **Information-theoretically secure** for the key-exchange step — no math, no Shor, no Grover.
- Limits: **dedicated dark fibre or DWDM wavelength**, ~120 km without trusted-node relays, dedicated hardware (Toshiba / ID Quantique).
- **Use selectively:** HQ–DC and inter-DC fibre inside a metro. **Not** for branches, customer apps, or DR > 200 km.

WHAT WE DO WHERE

Recommended option per zone

Zone	Recommended option	Why	Maturity
HQ Mumbai ↔ Primary DC Mumbai	Hybrid PQC + QKD at IPsec/MACsec layer; QKD AES-256 keys rotated every ~1 sec	Highest-value pipe; in metro QKD range; carries core-bank sync, HSM cluster, treasury	Production (HSBC, JPMC)
Primary DC Mumbai ↔ Far DR Delhi	PQC-only (Kyber+ECDH hybrid) over MACsec/IPsec; AES-256 rotated every 60 sec	1,150 km is beyond direct QKD; PQC + frequent rotation is correct answer for HNDL	NIST FIPS-203 (Aug 2024)
HQ ↔ Treasury / FX (intra-building)	MACsec + PQC firmware on switches; HSM-backed identity	Inside trust boundary; main concern is cert lifecycle and crypto-agility	Standard refresh
DC ↔ NPCI / RBI / SWIFT	PQC over carrier links; HSM-backed mutual TLS	Regulatory rails; can't self-deploy QKD; SWIFT CSP v2025/26 adds PQC	Vendor-led
Branches ↔ Regional Hub ↔ DC	PQC over SD-WAN/MPLS; TLS handshake = ECDH + Kyber hybrid	8,000+ sites; QKD physically infeasible; piggy-back on SD-WAN refresh	FIPS-203 standard
ATMs / POS	PQC libraries on next ATM refresh; HSM-backed PIN encryption stays AES	Slow refresh (~5–7 yrs); ride hardware refresh cycle	Vendor roadmap
Mobile / net banking	Hybrid TLS — X25519 + ML-KEM-768; HSM-backed signing keys → ML-DSA	Customer-facing; iOS 18, Chrome already negotiate Kyber	Production at FAANG scale

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How keys are stored in an HSM — the hierarchy

KEY HIERARCHY — NEVER LEAVES THE HSM IN CLEAR

LMK — Local Master Key (AES-256)

Generated in ceremony, split via Shamir, never appears in clear outside HSM

ZMK

Zone Master Key (per partner:
Visa, MC, NPCI, RBI)

KEK

Key Encryption Key
(intra-bank, intra-DC)

DEK roots

App-internal data
encryption roots

PIN keys, MAC keys, Card cryptograms (CVK, AC), DUKPT derivation
These actually encrypt customer PINs, sign card cryptograms

Working AES sessions for Finacle, fraud platform, middleware
Short-lived; rotated daily; used by app calls into HSM

Token / file encryption, Log integrity (HMAC), DB column encryption
App-layer protection; GDPR/RBI data localisation

Apps see only ***wrapped*** *key blobs from the database — the HSM unwraps inside its tamper-protected boundary, performs the op, discards plaintext.*

WHERE QUANTUM HITS — AND WHAT IS ALREADY SAFE

Crypto inside HSM	Quantum impact	Fix
AES-256 (LMK, KEK, working)	Grover halves to ~128-bit. OK	Stay on AES-256
3DES (legacy DUKPT, old PIN)	Broken classically already	Migrate to AES DUKPT
RSA-2048/3072 (TR-34, certs, code sign)	Shor breaks	ML-DSA / ML-KEM
ECDSA / ECDH (mTLS, key agreement)	Shor breaks	ML-KEM, ML-DSA
SHA-256 / SHA-3 (MAC, integrity)	Grover halves preimage. OK	Stay on SHA-256/384

Bottom line: payments are mostly AES-256 — already largely quantum-safe. Vulnerable parts are the ***public-key wrappers*** *around the keys. Those need PQC.*

Key ceremony and TR-31 / TR-34 transport

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LMK GENERATION — THE KEY CEREMONY

- **Multi-custodian**: 3–5 trusted officers (CISO, CRO, Head of Tech, Audit, RBI observer for top-tier).
- Officers gather in a secure room with **CCTV + signed witnesses**. HSM goes into ceremony mode via a physical key/smart card.
- HSM internally generates the LMK using its **TRNG** (avalanche-photodiode noise / quantum noise).
- LMK is **split via Shamir's Secret Sharing** — typically **3-of-5** or **2-of-3**: no single custodian can rebuild it; any quorum can.
- Each custodian receives their share on a **PIN-protected smart card** (Thales SafeNet / PSGcards) or in tamper-evident printed envelope.
- Ceremony log is signed and retained **10 years** (RBI / FIPS audit).
- To re-load the LMK (DR sync, firmware upgrade), the quorum reassembles, inserts cards, HSM reconstructs the LMK **internally**.
- **Why this matters**: protects the bank from any single insider walking out with all keys; this is also why LMK rotation is non-trivial — it triggers re-encryption of every wrapped key in the DB.

INTER-HSM TRANSPORT — TR-34 / TR-31 / PQ TR-34

- Need to move a ZMK between HDFC and NPCI / Visa / partner bank — you can't use the LMK because each org has its own.
- **TR-34 (one-time exchange)** — RSA digital envelope wraps the new ZMK + identity certificate. **This is the part vulnerable to Shor**.
- **TR-31 (key block format)** — once a shared KEK is in place, all subsequent keys move as AES-wrapped key blocks with CBC-MAC integrity. **Already quantum-safe**.
- **TR-34 PQ migration** — ANSI X9 working group has 2024 draft language replacing RSA wrap with **ML-KEM-1024**.
- Vendor support: **Thales payShield 10K firmware 2.x** ships ML-KEM/ML-DSA in 2024–25; **IBM CryptoExpress 8S (4770)** has PQC on z16.
- **Action for HDFC**: schedule next TR-34 ceremony with NPCI for FY27 once their PQ profile is published; in the meantime, all new ZMKs use 4096-bit RSA + AES-256 wrap (defence in depth).
- Existing TR-31 traffic (working keys flowing intra-bank, intra-DC) is **AES-wrapped** — no migration needed.

THE ARCHITECTURE

QKD keys flowing into HSM-anchored encryptors

HQ MUMBAI

5–30 km dark fibre

PRIMARY DC MUMBAI

5–30 km dark fibre

QKD photon channel — dedicated O-band wavelength — outputs identical 256-bit AES key every ~1 sec at both ends

QKD BOX (KME)

Toshiba MDI-QKD
or IDQ Clavis XGR

ETSI GS QKD 014/004 REST API delivers fresh AES-256 keys into the encryptor every ~1 second.

QKD BOX (KME)

Toshiba MDI-QKD
or IDQ Clavis XGR

ENCRYPTOR

Thales CN9100 / Senetas CN6140 / Ciena WL5e

ENCRYPTOR

Thales CN9100 / Senetas CN6140 / Ciena WL5e

AES-256 encrypted bulk traffic — 10–400 Gbps DWDM channels (core banking, card switch, treasury, HSM sync)

Encryptor authenticates to the HSM at boot (ML-DSA / ECDSA), receives its long-lived identity certificate.

HSM

Thales payShield 10K HSM cluster

HSM

Thales payShield 10K HSM cluster

CAPEX + OPEX ENVELOPE

Cost — Indian context, FY26 INR

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PROGRAMME CAPEX OVER 24–36 MONTHS

Item	Detail	Rs Cr
HSM refresh (Thales payShield 10K)	16 prod (DC) + 6 (DR) + 4 (HQ)	9–13
PQC integration services	Libraries + cert PKI + apps	6–10
QKD pair, HQ ↔ DC Mumbai	Toshiba MDI-QKD or IDQ Clavis	8–15
Quantum-safe encryptors	Thales CN9100 / Senetas CN6140	2.5–4
Dark fibre / DWDM wavelength capex	If not already owned (~30 km)	1–3
Branch-side PQC	TLS lib upgrade + cert refresh	4–6
PKI overhaul	Internal CA, OCSP, ML-DSA	3–5
CryptoBOM tooling	IBM Crypto Discovery / SandboxAQ	2–4
DR satellite-QKD R&D pilot	ISRO / TIFR partnership	5–8 (R&D)
Training, consulting, RBI engagement	Programme	3–5

ANNUAL OPEX (POST-ROLLOUT)

Item	Rs Cr / yr
HSM maintenance + FIPS recerts	1.5–2.5
QKD link maintenance	1.5–3
Dark fibre lease	1–2
PQC library subs (PQShield, etc.)	0.5–1
Annual key ceremonies + audits	0.5–1
Skilled crypto / quantum team (8–12 FTE)	4–6

THE NUMBER FOR THE MD

Rs 50–70 Cr

programme over FY26–FY28
plus Rs 10–15 Cr/yr opex

HOW WE GET THERE

Adoption roadmap — 36 months, 5 phases

Phase 0

Months 0–3

CryptoBOM

- Inventory **every** classical asymmetric crypto use (TLS, IPsec, code sign, certs, JWT, S/MIME, archived data).
- Tag by algorithm, key length, sensitivity, retention, replacement difficulty.
- Tools: IBM Crypto Discovery, SandboxAQ AQtive Guard, open-source crypto-cbom.
- Output: heat-map of where to migrate first.
- Cost: ~Rs 2–3 Cr.

Phase 1

Months 3–9

HSM + PKI foundation

- Refresh HSM fleet to current gen (Thales payShield 10K firmware ≥ 2.x, IBM CryptoExpress 8S).
- Both vendors ship **ML-KEM and ML-DSA** in 2024–25 firmware.
- Stand up post-quantum-ready internal CA (EJBCA 8.x with Dilithium, Entrust nShield CA).
- Add **crypto-agility wrapper** to internal libs so apps swap algos without code change.

Phase 2

Months 6–15

PQC pilot

- Pick one **non-customer-facing TLS endpoint** (admin portal, MIS dashboard).
- Deploy in **hybrid mode**: handshake = X25519 + ML-KEM-768.
- If ML-KEM is broken, X25519 still protects you; if X25519 is broken by quantum, ML-KEM does.
- Measure latency, cert size, library compat. Train SOC.

Phase 3

Months 12–24

PQC fleet rollout

- Roll PQC out to **all internal mTLS** between microservices.
- Migrate **branch ↔ regional hub IPsec** during SD-WAN refresh cycle.
- Migrate **mobile / net banking TLS** to hybrid (iOS 18, Chrome already negotiate Kyber).
- Upgrade **TR-34** ceremonies with NPCI / Visa / MC as their PQ profiles ratify.

Phase 4 + 5

Months 18–36+

QKD metro + DR R&D

- Procure **Toshiba MDI-QKD or IDQ Clavis XGR pair** for HQ ↔ Primary DC.
- Deploy Thales/Senetas/Ciena PQC encryptors on either side (ETSI GS QKD 014).
- Carry HSM cluster sync, treasury, Finacle replication. Shadow mode 6 mo, then prod.
- **DR R&D**: free-space / satellite QKD with TIFR / IISc / ISRO. NQM-funded.

DECISION REQUIRED

Call to action

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RECOMMENDATION

Approve **Phase 0 + Phase 1 funding of Rs 12–18 Cr for FY26**
— CryptoBOM inventory, HSM refresh to PQC-capable firmware, internal PKI overhaul.
Go / no-go gate at month 9 for **PQC pilot (Phase 2)**.
Full programme envelope **Rs 50–70 Cr** over FY26–FY28.

REGULATORY

RBI Master Direction on IT Governance (Nov 2023) and SEBI CSCRF (2024) flag PQC migration as a forward-looking control. NIST FIPS-203 / 204 / 205 finalised Aug 2024 — audit baseline now established.

THREAT REALITY

Harvest now, decrypt later is happening today on carrier fibre. KYC, loan files, treasury positions captured in 2024 will be readable in 2030–35. The clock is already running.

STRATEGIC

JPMC and HSBC have shipped QKD pilots; SBI, ICICI and Axis are early in PQC. **18-month head-start** maps directly to talent acquisition, IP, and India's National Quantum Mission positioning.

Author: SVP — Kranthi Molleti • HDFC Bank Quantum Computing Strategy